Jayachandra Sarika

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Recent CS graduate skilled in Python, SQL, Pandas, NumPy, and scikit-learn, with hands-on experience in data analysis, feature engineering, predictive modeling, and data pipelines. Experienced in translating complex datasets into actionable insights and scalable, data-driven solutions using cloud platforms, Git, and Agile practices.

EDUCATION

Master of Science, Computer Science 2024 - 2026
University of Houston, Houston TX GPA: 3.92 / 4.00

Bachelor of Technology, Computer Science and Engineering 2020 - 2024
Vellore Institute of Technology, Vellore India GPA: 3.64 / 4.00

TECHNICAL SKILLS

Programming & Data Science: Python (pandas, NumPy, scikit-learn, matplotlib, seaborn), R, SQL, Statistical Modeling, Machine Learning, Feature Engineering, Model Evaluation

Data Engineering & Pipelines: SQL (queries, joins, stored procedures, data extraction), Data Pipeline Development, ETL Workflows, Database Design, Data Quality Assurance, Automation (scripting, workflows)

Exploratory Data Analysis: Data Profiling, Missing Data Handling, Outlier Detection, Variance Analysis, Geospatial Analysis, Temporal Feature Engineering, Entity Matching Methods

Tools & Platforms: Power BI, Tableau, Excel (advanced: pivot tables, modeling, VBA), Version Control (Git, GitHub), Cloud Platforms (AWS, Azure, Databricks exposure), Agile/SDLC Practices

Business & Communication: Stakeholder Collaboration, Cross-Functional Teamwork, Technical Documentation, Dashboard Development, Business Value Quantification, ROI Analysis, Presentation Skills

WORK EXPERIENCE

Research Assistant August 2025 – May 2026
C.T. Bauer College of Business, University of Houston, Houston TX

- Supported faculty-led **marketing analytics** activities involving data cleaning, validation, **EDA**, visualization, and quantitative performance analysis across large-scale student datasets.
- Utilized **Python, Pandas, and Excel** to clean, transform, analyze, and validate structured datasets, identifying inconsistencies, outliers, and anomalous values.
- Analyzed quarterly **Sales Management Simulation (SMS)** decisions across industries of 10 competing companies, performing **anomaly detection, KPI analysis, and competitive benchmarking**.
- Evaluated business performance using **profit, market share, NPS, and company reputation**, analyzing relationships between strategic decisions, pricing, sales channels, and business outcomes.
- Developed **Power BI/Excel dashboards and visualizations** to communicate KPIs, performance trends, competitive comparisons, and data-driven insights.
- Communicated data-quality issues, performance trends, and analytical findings to faculty while applying concepts in **marketing analytics, consumer behavior, digital marketing, pricing, and quantitative research**.

Data Analyst Intern May 2023 – May 2024
BytesEdge Pvt. Ltd., Hyderabad, India

- Performed end-to-end **data cleaning and preprocessing** on structured datasets sourced from web and mobile platform operations using Python (Pandas, NumPy); handled missing value imputation, duplicate record removal, datatype standardization, and outlier detection to ensure data integrity prior to analysis.
- Wrote and optimized **SQL queries** (joins, subqueries, window functions, aggregations) to extract, transform, and load data from relational databases, supporting recurring business reporting needs and ad-hoc analytical requests from stakeholders.
- Designed and maintained interactive **business intelligence dashboards** in **Tableau** and **Microsoft Power BI**, visualizing KPIs such as user engagement metrics, platform performance trends, and client conversion rates to support data-driven decision-making across business units.
- Conducted **exploratory data analysis (EDA)** and **basic statistical analysis** - including descriptive statistics, correlation analysis, and trend identification - to surface actionable insights from client and operational datasets aligned with the company's digital agility and business intelligence focus.
- Built and automated **Excel-based reporting pipelines** using pivot tables, VLOOKUP/XLOOKUP, and conditional formatting to deliver weekly and monthly performance summaries to project leads, reducing manual reporting effort.
- Collaborated with cross-functional teams including software developers and project managers to translate business requirements into analytical frameworks, define data collection schemas, and ensure reporting outputs aligned with client deliverables.

PROJECTS

Forest Fire Detection - Image Classification & Real-Time Object Detection| *Python · PyTorch · YOLOv8 (Ultralytics) · torchvision · OpenCV · Scikit-learn · Streamlit* Aug 2026 – Sep 2026

- Designed and built an end-to-end wildfire detection pipeline spanning data acquisition, preprocessing, model training, evaluation, and deployment - fine-tuned a MobileNetV2 classifier (transfer learning) and trained a YOLOv8 object detector on 21,000+ annotated images (D-Fire dataset) to identify and localize fire/smoke via bounding boxes, demonstrating full-cycle ML ownership from prototype through a deployed application.
- Implemented a stratified 70/15/15 train/validation/test split, class-weighted loss function, and data augmentation (rotation, color jitter, random erasing) to correct 3:1 class imbalance and prevent overfitting, achieving 93.96% test accuracy / 96.46% fire recall on the classifier and 67.3% mAP@50 (72.1% precision) on the detector across 4,300+ held-out test images.
- Diagnosed and resolved a critical training-pipeline bug where a checkpoint-resume mechanism silently trained on an incorrect dataset, restoring correct 20-epoch training and improving validation mAP@50 from 64.5% to 67.1% through systematic debugging and iterative model refinement.
- Deployed a real-time inference system achieving ~90 FPS (11ms/frame) on CPU for live webcam/video fire monitoring, and built an interactive Streamlit web application enabling adjustable-confidence-threshold inference for both models with no code required

Gender Classification from Blogpost Text | *Python · Scikit-learn · TF-IDF · POS Tagging PCA · CNN · BERT/Contextual Embeddings · XGBoost* Jan 2025 – May 2025

- Designed a multi-stage NLP classification pipeline on a 3,232-entry labeled blog corpus, progressing from lexical baselines using TF-IDF and CountVectorizer representations with Logistic Regression (L1/L2), SVM, and XGBoost classifiers, to syntactically enriched feature sets via Part-of-Speech (POS) tag frequency vectors and POS-stratified TF-IDF applied independently across noun, verb, adjective, and adverb sub-corpora.
- Engineered handcrafted linguistic features - including sentiment polarity, subjectivity scores, noun/verb/adjective counts, word count, and average word length - combined with TF-IDF vectors and applied Principal Component Analysis (PCA) for dimensionality reduction to mitigate the curse of dimensionality and improve generalization across Logistic Regression, SVM, Random Forest, and Naive Bayes classifiers.
- Leveraged pre-trained contextual embeddings fed into a Convolutional Neural Network (CNN) for local semantic pattern extraction, demonstrating statistically significant accuracy gains over all traditional baselines and validating the superiority of deep contextual representations over handcrafted feature engineering for stylometric gender classification.

Lung Cancer Detection From CT Scans | *Python · PyTorch · MONAI · SimpleITK · Pandas · Scikit-learn · Jupyter* Jan 2024 – May 2024

- Built an end-to-end lung nodule detection pipeline on real CT scans from the LUNA16 dataset (178 scans), including volumetric image ingestion, Hounsfield Unit windowing, isotropic resampling, and lung segmentation via thresholding and connected-component analysis; implemented a self-contained candidate-detection stage using local-maxima search to locate nodules directly from full scans rather than relying on pre-labeled candidates.
- Trained and benchmarked 3D CNN classifiers - a custom lightweight CNN versus MONAI's DenseNet121 - on labeled nodule/false-positive patches, applying class-weighted loss, negative subsampling, and 3D flip augmentation to address a ~400:1 class imbalance; the DenseNet3D model improved test AUC from 0.83 to 0.96 and nodule-class precision from 37% to 73% at matched ~80% recall.
- Evaluated the full detection pipeline using the clinical-standard FROC methodology (sensitivity vs. false-positives-per-scan) on a held-out test split, computing the LUNA16 CPM score across seven standard operating points; prioritized sensitivity over precision consistent with a cancer-screening use case, and identified concrete data/segmentation improvements to close the gap toward competitive benchmarks.

CERTIFICATIONS

Applied Data Science powered by Google, *SmartInternz* May 2023 – July 2023
Google Cloud Computing Foundations , *Google* May 2023 – July 2023
Overview of Space Science, *ISRO* Aug 2023 – Sept 2023